Car Dealership & Rental Management System

Streamlining Vehicle Sales and Rentals with an Integrated Online Solution

Car Dealership & Rental Management System

Software Requirements Document (SRD)

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1. System Overview

The Car Dealership & Rental Management System is a comprehensive web-based platform designed to manage the entire lifecycle of vehicle inventory. This includes car purchasing, repair tracking, stock management, vehicle documentation, sales (cash, installment, rental), customer management, notifications, alerts, and financial and employee management. The system aims to maintain organized records, automate notifications, track profits, and manage customer interactions efficiently.

2. Core Modules

2.1 Car Purchase Management

The system will allow recording of car purchase information from suppliers.

Features:

- Add new car purchase entry
- Supplier details
- Purchase price
- Purchase date
- Upload images showing car condition at the time of purchase
- Upload purchase documents

Data Fields:

- Car ID
- Supplier name

- Supplier contact details
- Purchase price
- Purchase date
- Car details (brand, model, year, engine number, chassis number)
- Images of car condition at purchase
- Notes

3. Repair & Maintenance Management

Tracking repairs after car purchase.

Features:

- Add repair records
- Add repair costs
- Upload before and after repair images
- Total repair cost calculation

Data Fields:

- Car ID
- Repair description
- Parts replaced
- Labor cost
- Repair cost
- Repair date
- Images after repair

4. Inventory / Stock Management

Maintain an updated stock list of all cars.

Features:

- Real-time car inventory
- Car availability status
- Upload photos of cars
- Search and filter cars

Status Types:

- In stock
- Under repair
- Reserved
- Sold
- Rented

5. Vehicle Documentation Management

Track legal documents related to each vehicle.

Documents:

- Insurance
- Road permit
- Registration card

Features:

- Add issue date
- Add expiry date
- Document upload
- Automatic alerts before expiration

Alert System:

- Alerts 30, 15, and 7 days before expiry
- Alerts sent via SMS, email, and dashboard notifications

6. Car Sales Management

Cars can be sold through cash purchase, installment sale, or rental system.

6.1 Cash Purchase

Features:

- Full payment
- Discount option
- Agent commission option
- Sales invoice generation

Data Fields:

- Sale price
- Cash discount
- Agent name
- Agent commission

6.2 Installment Sale (Kisti)

Features:

- Installment plan creation
- Monthly installment calculation
- Automatic installment schedule
- Late payment fine
- Payment reminders

Example:

- If car price = 1,200,000 BDT
- Installment period = 24 months
- Monthly installment automatically calculated

Rules:

- Minimum down payment required
- Car delivery condition based on payment percentage (e.g., after 20% or 30% payment)

Late Payment System:

- Automatic late fine
- Reminder messages

Notifications:

- Installment due reminder
- Installment received
- Late payment notice

6.3 Rental System

Features:

- Rental contract creation
- Customer verification
- Rental duration
- Rental cost tracking

Customer Profile Data:

- Full name
- Phone number
- Address
- National ID
- Driving license
- Photo upload
- Emergency contact

Rental Notifications:

- Rental confirmation message
- Rental expiry reminder

7. Customer Management System

Maintain customer profiles.

Features:

- Create customer accounts
- View customer history
- Track purchases or rentals
- Document storage

Customer Records:

- Purchase history
- Rental history
- Installment records
- Outstanding payments

8. Automated Messaging System

Automatically send messages.

Message Types:

- Sales: Thank you message after purchase
- Installments: Payment reminder, Payment confirmation, Late payment notice
- Rental: Rental confirmation, Rental reminder

Messages sent via:

- SMS
- Email
- WhatsApp integration (optional)

9. Accounting & Financial Management

Calculate financial performance.

Expense Tracking:

- Purchase cost
- Repair cost
- Employee salary
- Other operational expenses

Income Tracking:

- Car sales
- Installment payments
- Rental income

Reports:

- Profit per car

- Monthly profit
- Sales report
- Expense report

10. Employee & Salary Management

Features:

- Employee profile management
- Salary structure
- Payment records

Data Fields:

- Employee name
- Role
- Salary amount
- Payment history
- Bonus/commission

11. Dashboard & Reporting

Dashboard Overview:

- Total cars in stock
- Cars sold
- Cars rented
- Pending installments
- Upcoming document expiry
- Total revenue
- Total profit

Reports:

- Car inventory report
- Sales report
- Rental report
- Installment report
- Profit and loss report

12. User Roles & Access Control

Admin:

- Full access to all modules

Manager:

- Inventory
- Sales

- Rental management

Accounts Officer:

- Financial management
- Salary management

Sales Agent:

- Customer management
- Sales records

13. Security & Data Management

Ensure:

- Secure login system
- Role-based access control
- Data backup system
- Encrypted sensitive information
- Activity logs for system users

14. Technical Requirements

Platform:

- Web-based application

Optional Extensions:

- Mobile app
- SMS gateway integration
- WhatsApp API integration
- Cloud storage for images and documents